



US 20250276257A1

(19) **United States**(12) **Patent Application Publication**
KARLES et al.(10) **Pub. No.: US 2025/0276257 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **E-VAPING DEVICE CARTRIDGE WITH
SUPERABSORBENT POLYMER**(71) Applicant: **Altria Client Services LLC**,
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Richmond, VA (US)(21) Appl. No.: **19/208,689**(22) Filed: **May 15, 2025****Related U.S. Application Data**

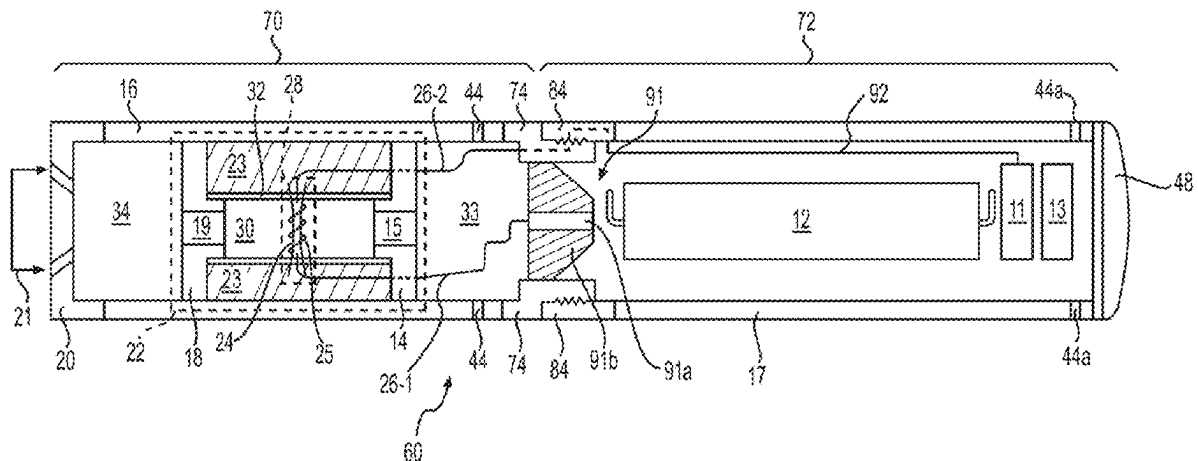
(63) Continuation of application No. 18/501,587, filed on Nov. 3, 2023, now Pat. No. 12,311,288, which is a continuation of application No. 17/734,488, filed on May 2, 2022, now Pat. No. 11,826,675, which is a continuation of application No. 16/413,977, filed on May 16, 2019, now Pat. No. 11,338,223, which is a continuation of application No. 15/192,136, filed on Jun. 24, 2016, now Pat. No. 10,292,426.

Publication Classification(51) **Int. Cl.**
B01D 15/22 (2006.01)
A24F 40/10 (2020.01)
A24F 40/42 (2020.01)
A24F 40/70 (2020.01)
B01J 20/26 (2006.01)**B05D 5/00** (2006.01)**B05D 7/22** (2006.01)**B05D 7/24** (2006.01)**B65D 81/26** (2006.01)**D03D 15/00** (2021.01)**D03D 15/283** (2021.01)**D03D 15/50** (2021.01)(52) **U.S. Cl.**CPC **B01D 15/22** (2013.01); **A24F 40/42**(2020.01); **B01J 20/267** (2013.01); **B05D 5/00**(2013.01); **B05D 7/227** (2013.01); **B05D 7/24**(2013.01); **B65D 81/264** (2013.01); **D03D****15/00** (2013.01); **D03D 15/283** (2021.01);**D03D 15/50** (2021.01); **A24F 40/10** (2020.01);**A24F 40/70** (2020.01)

(57)

ABSTRACT

A cartridge for an e-vaping device includes a reservoir configured to hold pre-vapor formulation, a vaporizer assembly configured to draw at least some of the pre-vapor formulation from the reservoir and vaporize the drawn pre-vapor formulation to form a vapor, and a superabsorbent polymer configured to absorb free water from pre-vapor formulation held in the reservoir. The superabsorbent polymer includes a cross-linked polyacrylate copolymer that is substantially inert to the pre-vapor formulation. The superabsorbent polymer may be included in a layer on one or more surfaces in the cartridge. The layer may include the superabsorbent polymer and a binder. The superabsorbent polymer may be included with the pre-vapor formulation in a formulation mixture. The superabsorbent polymer may be included in an interior of at least one element comprising the cartridge. The superabsorbent polymer may be included in a separate compartment in the reservoir.



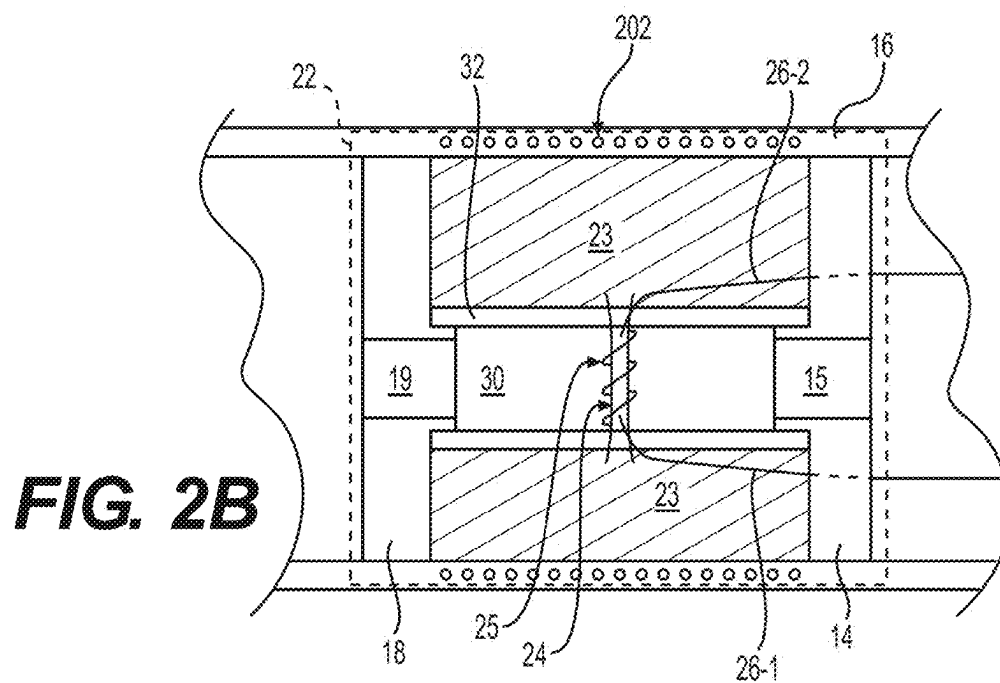
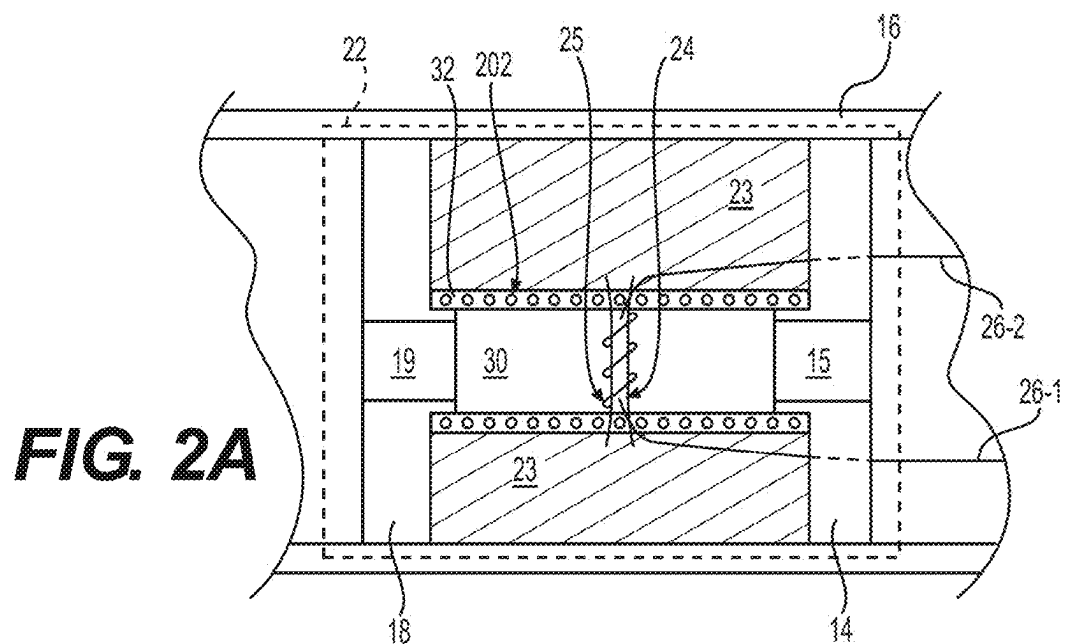


FIG. 2C

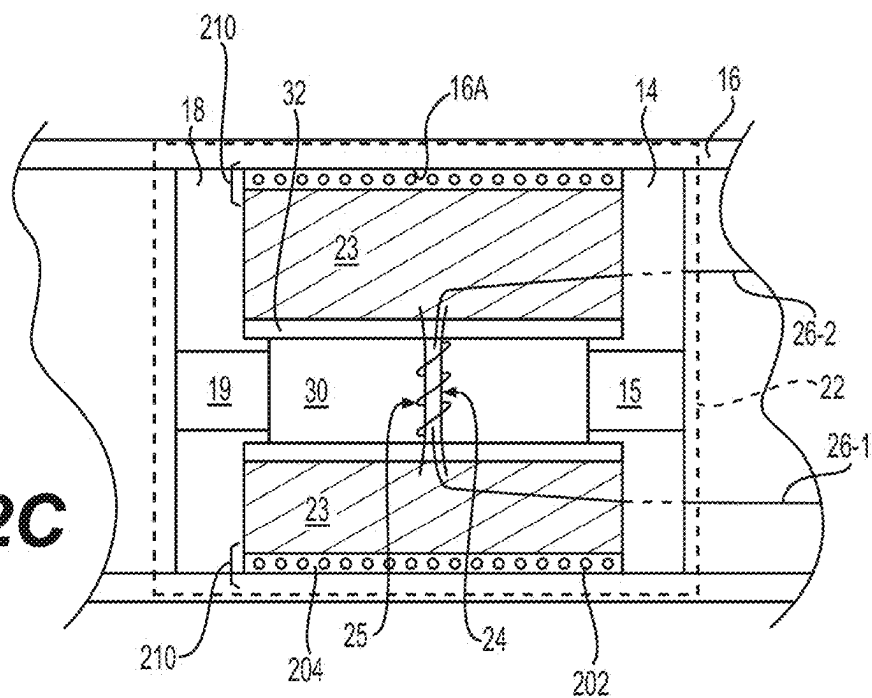
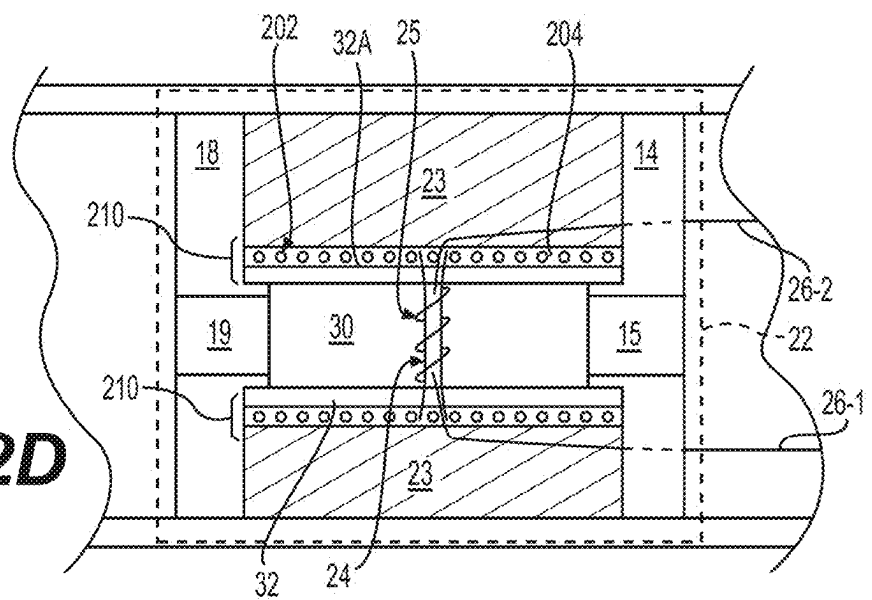
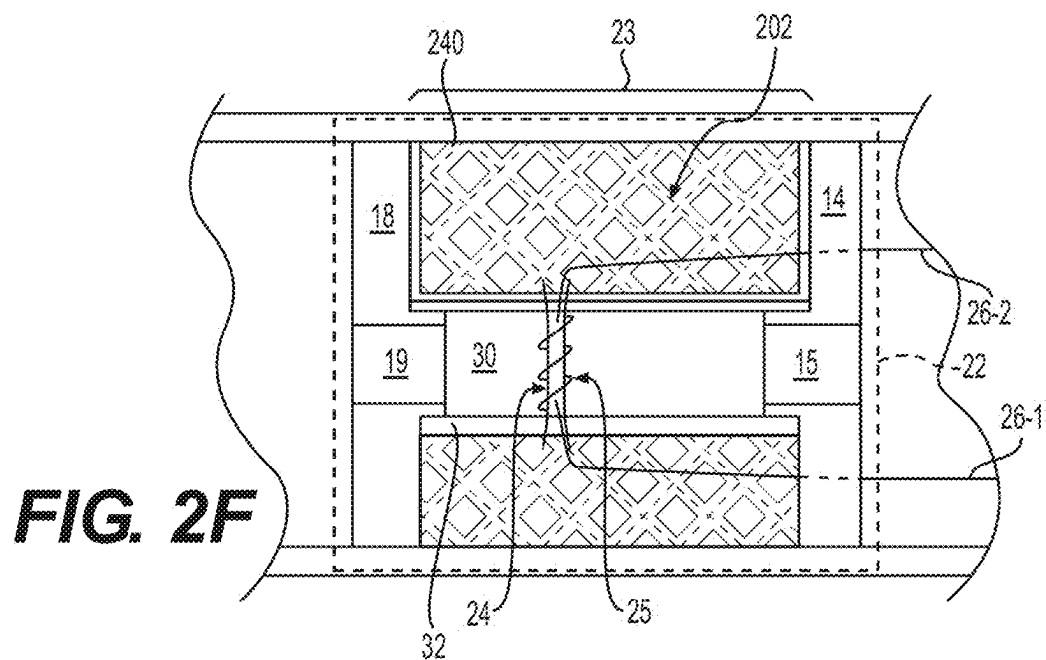
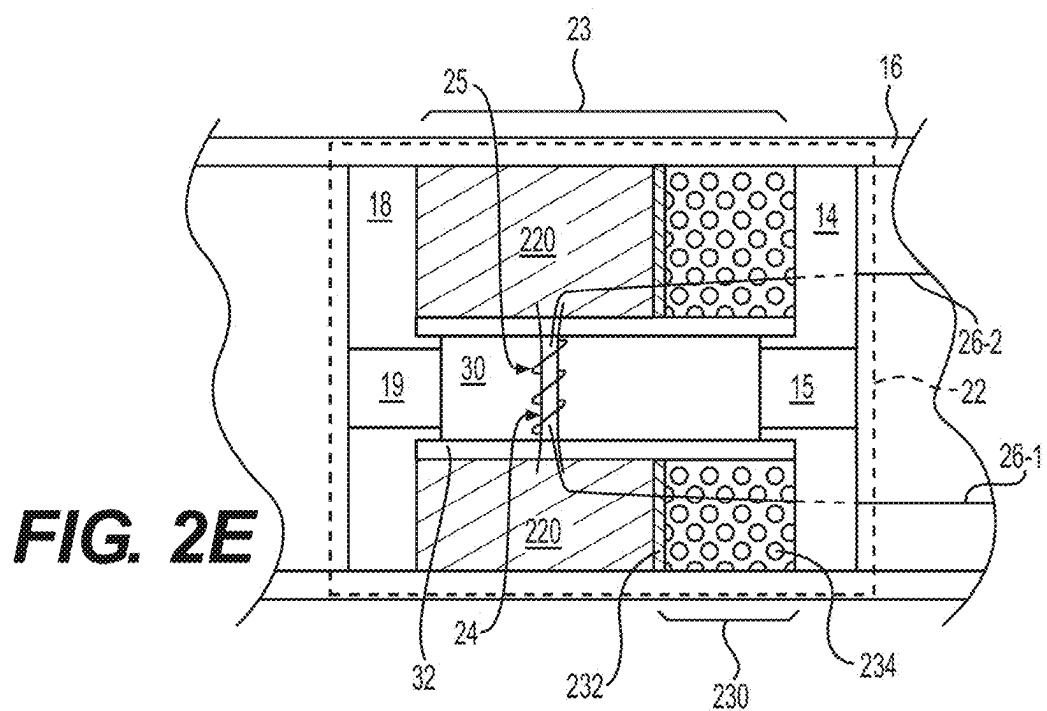


FIG. 2D





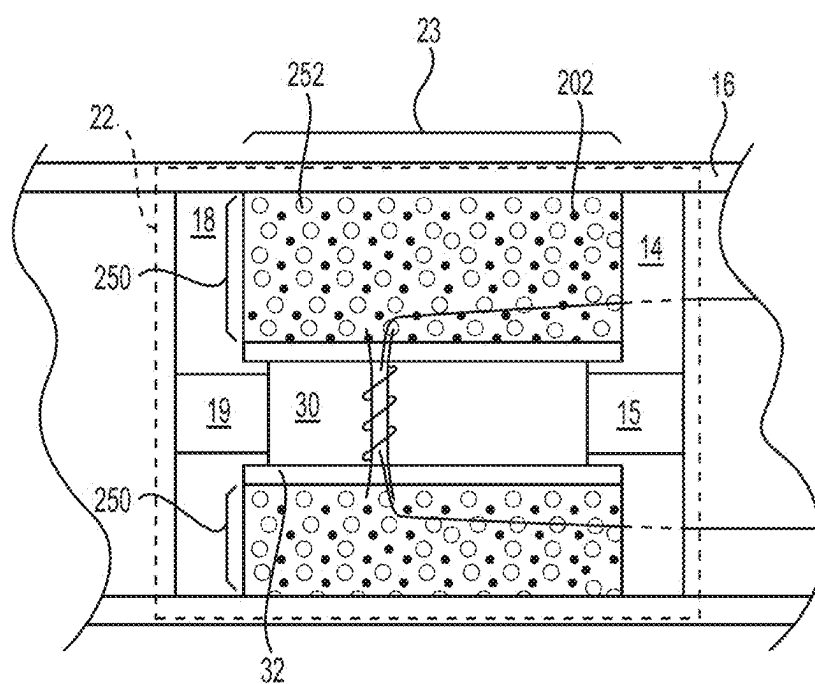


FIG. 2G

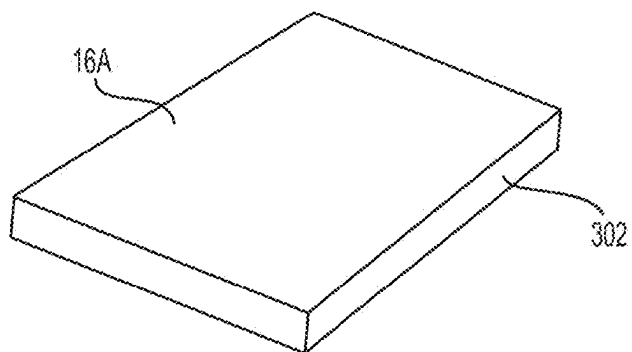


FIG. 3A

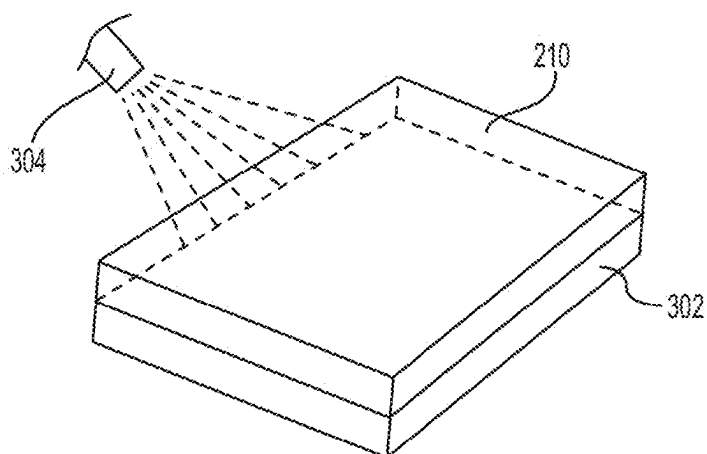


FIG. 3B

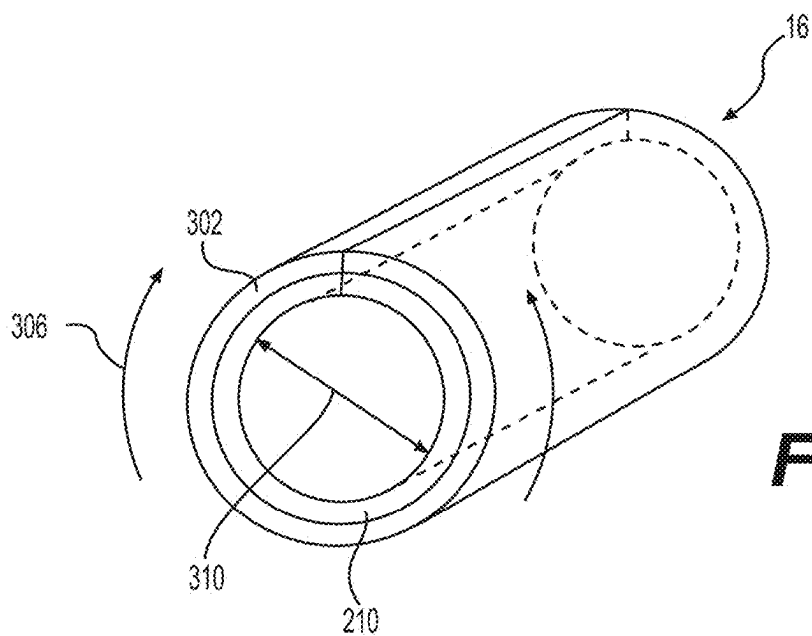


FIG. 3C

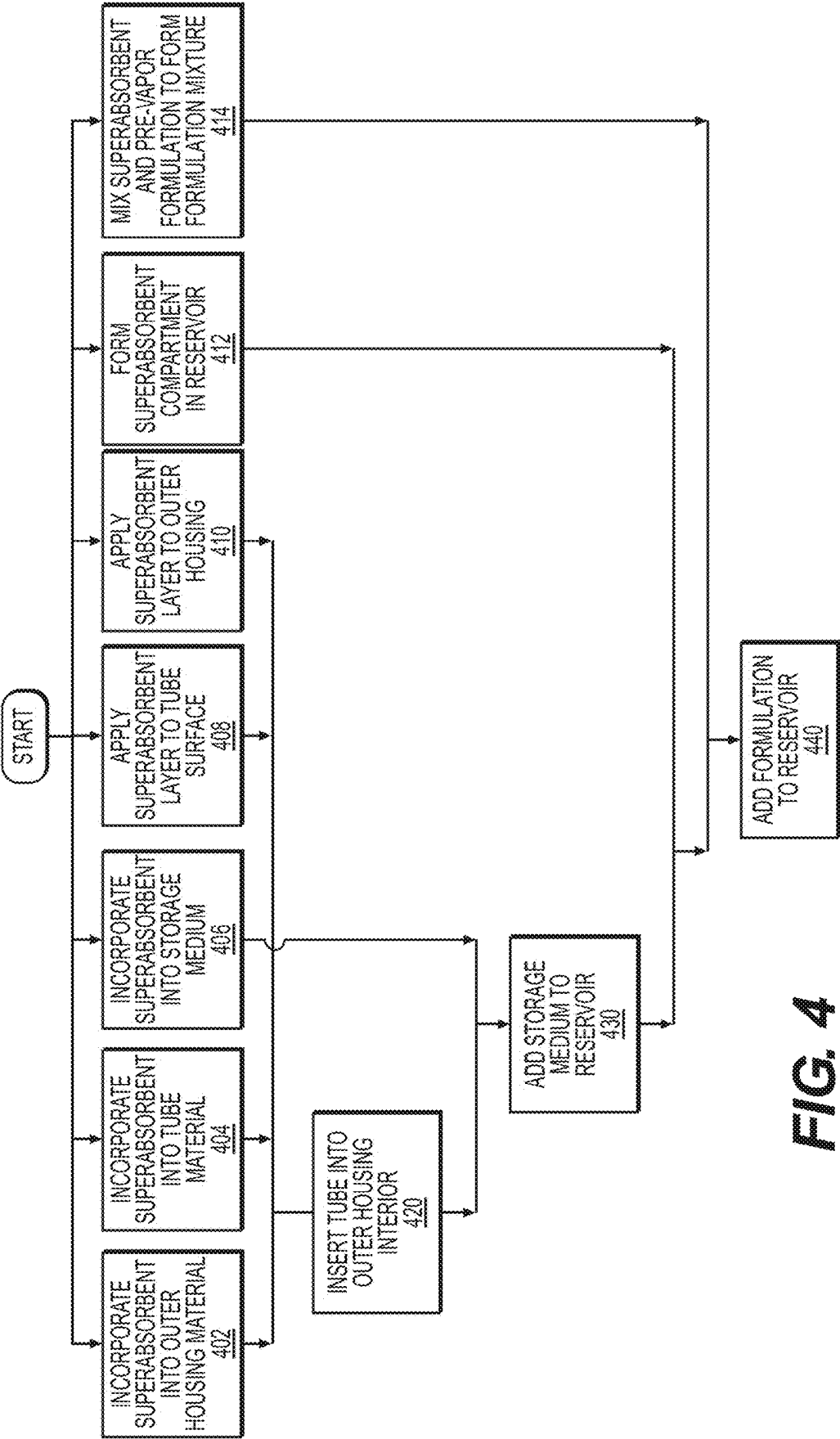


FIG. 4

E-VAPING DEVICE CARTRIDGE WITH SUPERABSORBENT POLYMER

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application is a continuation application of U.S. application Ser. No. 18/501,587, filed on Nov. 3, 2023, which is a continuation of U.S. application Ser. No. 17/734,488, filed on May 2, 2022, which is a continuation application of U.S. application Ser. No. 16/413,977, filed on May 16, 2019, which is a continuation application of U.S. application Ser. No. 15/192,136, filed Jun. 24, 2016, the disclosures of each of which are incorporated by reference in their entireties.

BACKGROUND

Field

[0002] One or more example embodiments relate to electronic vaping and/or e-vaping devices.

Description of Related Art

[0003] E-vaping devices, also referred to herein as electronic vaping devices (EVDs) may be used by adult vapers for portable vaping. An e-vaping device may vaporize a pre-vapor formulation to form a vapor. The e-vaping device may include a reservoir that holds a pre-vapor formulation and a heating element that vaporizes the pre-vapor formulation by applying heat to at least a portion of the pre-vapor formulation.

[0004] In some cases, one or more elements in an e-vaping device may be a metallic element. A metallic element may be in fluid communication with a pre-vapor formulation in the e-vaping device.

[0005] In some cases, a metallic element in an e-vaping device may be at least partially corroded based on being in fluid communication with the pre-vapor formulation. Such corrosion may reduce an operational lifetime of the metallic element and/or may reduce a sensory experience provided by the e-vaping device.

SUMMARY

[0006] According to some example embodiments, a cartridge for an e-vaping device may include a reservoir configured to hold a pre-vapor formulation and a vaporizer assembly configured to draw at least some of the pre-vapor formulation from the reservoir. The reservoir may include a superabsorbent polymer in fluid communication with an interior of the reservoir. The superabsorbent polymer may be configured to absorb free water from the pre-vapor formulation held in the reservoir. The superabsorbent polymer may include a cross-linked polyacrylate copolymer that is substantially inert to the pre-vapor formulation. The vaporizer assembly may be configured to draw at least some of the pre-vapor formulation from the reservoir. The vaporizer assembly may be further configured to vaporize the drawn pre-vapor formulation to form a vapor.

[0007] The reservoir may include a superabsorbent layer on at least one surface, the at least one surface at least partially defining the interior of the reservoir. The superabsorbent layer may include the superabsorbent polymer and at least one binder.

[0008] The reservoir may include a storage medium configured to hold the pre-vapor formulation. The superabsorbent polymer may be included in the storage medium.

[0009] The reservoir may include a cylindrical outer housing at least partially defining an outer surface of the reservoir. An inner tube may define an inner surface of the reservoir, such that the reservoir interior is an annulus. A superabsorbent layer may be on at least one of the inner surface of the outer housing and the outer surface of the inner tube. The superabsorbent layer may include the superabsorbent polymer and at least one binder.

[0010] The superabsorbent polymer may be included within an interior of at least one of the outer housing and the inner tube.

[0011] The reservoir may include a formulation compartment and a superabsorbent compartment. The formulation compartment may be configured to hold the pre-vapor formulation. The superabsorbent compartment may include the superabsorbent polymer and a selectively-permeable partition. The partition may be configured to enable pre-vapor circulation between the formulation compartment and the superabsorbent compartment, and restrict the superabsorbent polymer to the superabsorbent compartment.

[0012] The selectively permeable partition may enclose the superabsorbent polymer within the formulation compartment.

[0013] According to some example embodiments, an e-vaping device may include a cartridge and a power supply. The cartridge may include a reservoir configured to hold a pre-vapor formulation and a vaporizer assembly configured to draw at least some of the pre-vapor formulation from the reservoir. The reservoir may include a superabsorbent polymer in fluid communication with an interior of the reservoir. The superabsorbent polymer may be configured to absorb free water from the pre-vapor formulation held in the reservoir the superabsorbent polymer including a cross-linked polyacrylate copolymer that is substantially inert to the pre-vapor formulation. The vaporizer assembly may be configured to draw at least some of the pre-vapor formulation from the reservoir. The vaporizer assembly may be further configured to vaporize the drawn pre-vapor formulation to form a vapor. The power supply may be configured to supply electrical power to the vaporizer assembly.

[0014] The reservoir may include a superabsorbent layer on at least one surface. The at least one surface may at least partially define the interior of the reservoir. The superabsorbent layer may include the superabsorbent polymer and at least one binder.

[0015] The reservoir may include a storage medium configured to hold the pre-vapor formulation. The superabsorbent polymer may be included in the storage medium.

[0016] The reservoir may include a cylindrical outer housing at least partially defining an outer surface of the reservoir, an inner tube defining an inner surface of the reservoir, such that the reservoir interior is an annulus, and a superabsorbent layer on at least one of the inner surface of the outer housing and the outer surface of the inner tube, the superabsorbent layer including the superabsorbent polymer and at least one binder.

[0017] The superabsorbent polymer may be included within an interior of at least one of the outer housing and the inner tube.

[0018] The reservoir may include a formulation compartment and a superabsorbent compartment, the formulation

compartment configured to hold the pre-vapor formulation, the superabsorbent compartment including the superabsorbent polymer and a selectively-permeable partition. The partition may be configured to enable pre-vapor circulation between the formulation compartment and the superabsorbent compartment, and restrict the superabsorbent polymer to the superabsorbent compartment.

[0019] The selectively permeable partition may enclose the superabsorbent polymer within the formulation compartment.

[0020] The power supply may include a rechargeable battery.

[0021] The cartridge and the power supply may be removably coupled together.

[0022] According to some example embodiments, a cartridge for an e-vaping device may include a reservoir holding a formulation mixture and a vaporizer assembly configured to draw at least some of the pre-vapor formulation from the reservoir. The formulation mixture may include a pre-vapor formulation, and a superabsorbent polymer configured to absorb free water from the pre-vapor formulation. The superabsorbent polymer may include a cross-linked polyacrylate copolymer that is substantially inert to the pre-vapor formulation. The vaporizer assembly may be configured to draw at least some of the pre-vapor formulation from the reservoir. The vaporizer assembly may be further configured to vaporize the drawn pre-vapor formulation to form a vapor.

[0023] According to some example embodiments, a method may include configuring a cartridge to form a vapor based on vaporization of a substantially de-watered pre-vapor formulation. The cartridge may include a reservoir configured to hold a pre-vapor formulation and a vaporizer assembly configured to vaporize pre-vapor formulation drawn from the reservoir. The configuring may include applying a superabsorbent polymer to at least one surface configured to be in fluid communication with an interior of the reservoir. The superabsorbent polymer may be configured to absorb free water from the pre-vapor formulation. The superabsorbent polymer may include a cross-linked polyacrylate copolymer that is substantially inert to the pre-vapor formulation.

[0024] The reservoir may include a storage medium configured to hold the pre-vapor formulation within the interior of the reservoir, and applying the superabsorbent polymer includes applying the superabsorbent polymer to the storage medium.

[0025] The reservoir may include a cylindrical outer housing defining an outer surface of the reservoir and an inner tube defining an inner surface of the reservoir, such that the reservoir interior is an annulus. Applying the superabsorbent polymer includes applying a superabsorbent layer on at least one of the inner surface of the outer housing and the outer surface of the inner tube. The superabsorbent layer may include the superabsorbent polymer and at least one binder.

[0026] The superabsorbent polymer may include a fibrous material. Applying the superabsorbent polymer may include weaving the superabsorbent polymer into the at least one surface. Applying the superabsorbent polymer may include impregnating the superabsorbent polymer into an interior of the at least one surface.

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] The various features and advantages of the non-limiting embodiments described herein become more apparent upon review of the detailed description in conjunction with the accompanying drawings. The accompanying drawings are merely provided for illustrative purposes and should not be interpreted to limit the scope of the claims. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. For purposes of clarity, various dimensions of the drawings may have been exaggerated.

[0028] FIG. 1A is a side view of an e-vaping device according to some example embodiments.

[0029] FIG. 1B is a cross-sectional view along line IB-IB' of the e-vaping device of FIG. 1A.

[0030] FIG. 2A is a cross-sectional view of a vapor generator that includes a superabsorbent polymer, according to some example embodiments.

[0031] FIG. 2B is a cross-sectional view of a vapor generator that includes a superabsorbent polymer, according to some example embodiments.

[0032] FIG. 2C is a cross-sectional view of a vapor generator that includes a superabsorbent layer, according to some example embodiments.

[0033] FIG. 2D is a cross-sectional view of a vapor generator that includes a superabsorbent layer, according to some example embodiments.

[0034] FIG. 2E is a cross-sectional view of a vapor generator that includes a superabsorbent compartment, according to some example embodiments.

[0035] FIG. 2F is a cross-sectional view of a vapor generator that includes a superabsorbent polymer within a storage medium in the reservoir, according to some example embodiments.

[0036] FIG. 2G is a cross-sectional view of a vapor generator that includes a reservoir holding a formulation mixture of pre-vapor formulation and superabsorbent polymer, according to some example embodiments.

[0037] FIG. 3A, FIG. 3B, and FIG. 3C illustrate a method of at least partially forming a cartridge that includes a superabsorbent polymer on a surface that at least partially defines an interior of a reservoir of the cartridge, according to some example embodiments.

[0038] FIG. 4 is a flowchart illustrating a method for configuring a cartridge to absorb free water from a pre-vapor formulation held in the reservoir, according to some example embodiments.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

[0039] Some detailed example embodiments are disclosed herein. However, specific structural and functional details disclosed herein are merely representative for purposes of describing example embodiments. Example embodiments may, however, be embodied in many alternate forms and should not be construed as limited to only the example embodiments set forth herein.

[0040] Accordingly, while example embodiments are capable of various modifications and alternative forms, example embodiments thereof are shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that there is no intent to limit example embodiments to the particular forms dis-

closed, but to the contrary, example embodiments are to cover all modifications, equivalents, and alternatives falling within the scope of example embodiments. Like numbers refer to like elements throughout the description of the figures.

[0041] It should be understood that when an element or layer is referred to as being “on,” “connected to,” “coupled to,” or “covering” another element or layer, it may be directly on, connected to, coupled to, or covering the other element or layer or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly connected to,” or “directly coupled to” another element or layer, there are no intervening elements or layers present. Like numbers refer to like elements throughout the specification. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0042] It should be understood that, although the terms first, second, third, etc. may be used herein to describe various elements, regions, layers and/or sections, these elements, regions, layers, and/or sections should not be limited by these terms. These terms are only used to distinguish one element, region, layer, or section from another region, layer, or section. Thus, a first element, region, layer, or section discussed below could be termed a second element, region, layer, or section without departing from the teachings of example embodiments.

[0043] Spatially relative terms (e.g., “beneath,” “below,” “lower,” “above,” “upper,” and the like) may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It should be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the term “below” may encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

[0044] The terminology used herein is for the purpose of describing various example embodiments only and is not intended to be limiting of example embodiments. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, and/or elements, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, and/or groups thereof.

[0045] Example embodiments are described herein with reference to cross-sectional illustrations that are schematic illustrations of idealized embodiments (and intermediate structures) of example embodiments. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, example embodiments should not be construed as limited to the shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.

[0046] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which example embodiments belong. It will be further understood that terms, including those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0047] FIG. 1A is a side view of an e-vaping device **60** according to some example embodiments. FIG. 1B is a cross-sectional view along line IB-IB' of the e-vaping device of FIG. 1A. The e-vaping device **60** may include one or more of the features set forth in U.S. Patent Application Publication No. 2013/0192623 to Tucker et al. filed Jan. 31, 2013 and U.S. Patent Application Publication No. 2013/0192619 to Tucker et al. filed Jan. 14, 2013, the entire contents of each of which are incorporated herein by reference thereto. As used herein, the term “e-vaping device” is inclusive of all types of electronic vaping devices, regardless of form, size or shape.

[0048] Referring to FIG. 1A and FIG. 1B, an e-vaping device **60** includes a replaceable cartridge (or first section) **70** and a reusable power supply section (or second section) **72**. Sections **70**, **72** are removably coupled together at complimentary interfaces **74**, **84** of the respective cartridge **70** and power supply section **72**.

[0049] In some example embodiments, the interfaces **74**, **84** are threaded connectors. It should be appreciated that each interface **74**, **84** may be any type of connector, including a snug-fit, detent, clamp, bayonet, and/or clasp. One or more of the interfaces **74**, **84** may include a cathode connector, anode connector, some combination thereof, etc. to electrically couple one or more elements of the cartridge **70** to one or more power supplies **12** in the power supply section **72** when the interfaces **74**, **84** are coupled together.

[0050] As shown in FIG. 1A and FIG. 1B, in some example embodiments, an outlet end insert **20** is positioned at an outlet end of the cartridge **70**. The outlet end insert **20** includes at least one outlet port **21** that may be located off-axis from the longitudinal axis of the e-vaping device **60**. The at least one outlet port **21** may be angled outwardly in relation to the longitudinal axis of the e-vaping device **60**. Multiple outlet ports **21** may be uniformly or substantially uniformly distributed about the perimeter of the outlet end insert **20** so as to uniformly or substantially uniformly distribute a vapor drawn through the outlet end insert **20** during vaping. Thus, as a vapor is drawn through the outlet end insert **20**, the vapor may move in different directions.

[0051] The cartridge **70** includes a vapor generator **22**. The vapor generator **22** includes at least a portion of an outer housing **16** of the cartridge **70** extending in a longitudinal direction and an inner tube (or chimney) **32** coaxially positioned within the outer housing **16**. The power supply section **72** includes an outer housing **17** extending in a longitudinal direction. In some example embodiments, the outer housing **16** may be a single tube housing both the cartridge **70** and the power supply section **72**. In the example embodiment illustrated in FIG. 1A and FIG. 1B, the entire e-vaping device **60** may be disposable.

[0052] The outer housings **16**, **17** may each have a generally cylindrical cross-section. In some example embodiments, the outer housings **16**, **17** may each have a generally triangular cross-section along one or more of the cartridge

70 and the power supply section 72. In some example embodiments, the outer housing 17 may have a greater circumference or dimensions at a tip end than a circumference or dimensions of the outer housing 16 at an outlet end of the e-vaping device 60.

[0053] At one end of the inner tube 32, a nose portion of a gasket (or seal) 14 is fitted into an end portion of the inner tube 32. An outer perimeter of the gasket 14 provides a substantially airtight seal with an interior surface of the outer housing 16. The gasket 14 includes a channel 15. The channel 15 opens into an interior of the inner tube 32 that defines a central channel 30. A space 33 at a backside portion of the gasket 14 assures communication between the channel 15 and one or more air inlet ports 44. Air may be drawn into the space 33 in the cartridge 70 through the one or more air inlet ports 44 during vaping, and the channel 15 may enable such air to be drawn into the central channel 30 of the vapor generator 22.

[0054] In some example embodiments, a nose portion of another gasket 18 is fitted into another end portion of the inner tube 32. An outer perimeter of the gasket 18 provides a substantially tight seal with an interior surface of the outer housing 16. The gasket 18 includes a channel 19 disposed between the central channel 30 of the inner tube 32 and a space 34 at an outlet end of the outer housing 16. The channel 19 may transport a vapor from the central channel 30 to exit the vapor generator 22 to the space 34. The vapor may exit the cartridge 70 from space 34 through the outlet insert 20.

[0055] In some example embodiments, at least one air inlet port 44 is formed in the outer housing 16, adjacent to the interface 74 to reduce and/or minimize the chance of an adult vaper's fingers occluding one of the ports and to control the resistance-to-draw (RTD) during vaping. In some example embodiments, the air inlet ports 44 may be machined into the outer housing 16 with precision tooling such that their diameters are closely controlled and replicated from one e-vaping device 60 to the next during manufacture.

[0056] In a further example embodiment, the air inlet ports 44 may be drilled with carbide drill bits or other high-precision tools and/or techniques. In yet a further example embodiment, the outer housing 16 may be formed of metal or metal alloys such that the size and shape of the air inlet ports 44 may not be altered during manufacturing operations, packaging, and/or vaping. Thus, the air inlet ports 44 may provide more consistent RTD. In yet a further example embodiment, the air inlet ports 44 may be sized and configured such that the e-vaping device 60 has a RTD in the range of from about 60 mm H₂O to about 150 mm H₂O.

[0057] Still referring to FIG. 1A and FIG. 1B, the vapor generator 22 includes a reservoir 23. The reservoir 23 is configured to hold one or more pre-vapor formulations. The reservoir 23 is contained in an outer annulus between the inner tube 32 and the outer housing 16 and between the gaskets 14 and 18. Thus, the reservoir 23 at least partially surrounds the central channel 30. The reservoir 23 may include a storage medium configured to store the pre-vapor formulation therein. A storage medium included in a reservoir 23 may include a winding of cotton gauze or other fibrous material about a portion of the cartridge 70.

[0058] In some example embodiments, the reservoir 23 is configured to hold different pre-vapor formulations. For example, the reservoir 23 may include one or more sets of

storage media, where the one or more sets of storage media are configured to hold different pre-vapor formulations.

[0059] A pre-vapor formulation, as described herein, is a material or combination of materials that may be transformed into a vapor. For example, the pre-vapor formulation may be a liquid, solid and/or gel formulation including, but not limited to, water, beads, solvents, active ingredients, ethanol, plant extracts, natural or artificial flavors, and/or pre-vapor formulation such as glycerin and propylene glycol. Different pre-vapor formulations may include different elements. Different pre-vapor formulations may have different properties. For example, different pre-vapor formulations may have different viscosities when the different pre-vapor formulations are at a common temperature. One or more of pre-vapor formulations may include those described in U.S. Patent Application Publication No. 2015/0020823 to Lipowicz et al. filed Jul. 16, 2014 and U.S. Patent Application Publication No. 2015/0313275 to Anderson et al. filed Jan. 21, 2015, the entire contents of each of which is incorporated herein by reference thereto.

[0060] Still referring to FIG. 1A and FIG. 1B, in some example embodiments, the reservoir 23 includes a superabsorbent polymer in fluid communication with an interior of the reservoir 23. The superabsorbent polymer may be in fluid communication with the pre-vapor formulation in the reservoir 23. The superabsorbent polymer may be substantially inert to the pre-vapor formulation.

[0061] The superabsorbent polymer may absorb one or more fluids, including one or more liquid materials. The one or more fluids may include one or more polar compound materials, including water, ethanol, some combination thereof, or the like. The superabsorbent polymer may absorb one or more fluids, including free water, from the pre-vapor formulation held in the reservoir 23. The pre-vapor formulation held in the reservoir 23 may be hygroscopic and may absorb an amount of water, from the atmosphere, that is equal to or greater than 10% of the mass of the pre-vapor formulation at saturation (100% relative humidity). The superabsorbent polymer may absorb such water (free water, "moisture," etc.) from the pre-vapor formulation.

[0062] The superabsorbent polymer, the one or more locations of the superabsorbent polymer relative to one or more portions of the reservoir 23, the absorbing of free water by the superabsorbent layer, some combination thereof, or the like are described in further detail below with reference to FIGS. 2A-G.

[0063] Based on including a superabsorbent polymer that may absorb free water from the pre-vapor formulation, the cartridge 70 may be configured to reduce and/or prevent corrosion of metallic elements of the cartridge 70 by free water. Such metallic elements may include, for example, at least one of the heating element 25 and the electrical leads 26-1 and 26-2. In addition, the cartridge 70 may be configured to reduce and/or prevent partitioning of one or more elements in a vapor formed in the cartridge 70 based on the presence of free water in the pre-vapor formulation. Therefore, the cartridge 70 may be configured to reduce and/or prevent a harshness of the sensory experience that may be provided by a vapor that includes such partitioning.

[0064] As described further below with reference to FIGS. 2A-G, the superabsorbent polymer may be included in the reservoir 23 in the form of a powder, as dispersed particles or fibers, as a layer on one or more surfaces in the cartridge 70, as a mixture with a material included in the cartridge 70,

as a part of a mixture with the pre-vapor formulation, as a fibrous material that is interwoven with a material comprising one or more elements of the cartridge 70, some combination thereof, or the like.

[0065] The superabsorbent polymer may be a cross-linked polyacrylate copolymer that may be safely used as an article or element of one or more articles configured for use in contact with food. The superabsorbent polymer may include one or more cross-linked polyacrylate copolymers. The cross-linked polyacrylate copolymers may include at least one of grafted copolymers of cross-linked sodium polyacrylate identified as 2-propenoic acid, polymers with N,N-di-2-propenyl-2-propen-1-amine and hydrolyzed polyvinyl acetate, sodium salts, graft (CAS Reg. No. 166164-74-5), and 2-propenoic acid, polymers with 2-ethyl-2-((1-oxo-2-propenyl)oxy)methyl)-1,3-propanediyl di-2-propenoate and sodium 2-propenoate (CAS Reg. No. 76774-25-9). The cross-linked polyacrylate copolymers may include one or more adjuvant substances used in the production of such copolymers. The one or more adjuvant substances may include one or more substances generally recognized as safe in food, and one or more substances used in accordance with a prior government sanction or approval. The cross-linked polyacrylate copolymers may yield low molecular weight (less than 1,000 Daltons) extractives of no more than 0.15 percent by weight of the total polymer when extracted with 0.2 percent by weight of aqueous sodium chloride solution at 20° C. for 24 hours. The low molecular weight extractives may be determined using size exclusion chromatography or an equivalent method.

[0066] Still referring to FIG. 1A and FIG. 1B, the vapor generator 22 includes a vaporizer assembly 28. The vaporizer assembly 28 is configured to vaporize at least a portion of the pre-vapor formulation held in the reservoir 23 to form a vapor.

[0067] The vaporizer assembly 28 includes a dispensing interface 24. The dispensing interface 24 may be coupled to the reservoir 23. The dispensing interface 24 is configured to draw one or more pre-vapor formulations from the reservoir 23. Pre-vapor formulation drawn from the reservoir 23 into the dispensing interface 24 may be drawn into an interior of the dispensing interface 24. It will be understood, therefore, that pre-vapor formulation drawn from a reservoir 23 into a dispensing interface 24 may include pre-vapor formulation held in the dispensing interface 24.

[0068] Still referring to FIG. 1A and FIG. 1B, the vaporizer assembly 28 further includes a heating element 25. The heating element 25 may be coupled to the dispensing interface 24. In some example embodiments, the heating element 25 may be directly coupled to the dispensing interface 24 such that the heating element 25 is coupled to an exterior surface of the dispensing interface 24. The heating element 25 may at least partially surround a portion of the dispensing interface 24 such that when the heating element 25 is activated, one or more pre-vapor formulations in the dispensing interface 24 may be vaporized by the heating element 25 to form a vapor. In some example embodiments, including the example embodiment illustrated in FIG. 1B, the heating element 25 completely surrounds the dispensing interface 24.

[0069] In some example embodiments, including the example embodiments shown in FIG. 1B, and as shown further with reference to FIG. 2A and FIG. 2B, the heating element 25 includes a heater coil wire that extends around

the exterior surface of the dispensing interface 24. The heating element 25 may heat one or more portions of the dispensing interface 24, including at least some of the pre-vapor formulation held in the dispensing interface 24, to vaporize the at least some of the pre-vapor formulation held in the dispensing interface 24.

[0070] The heating element 25 may heat one or more pre-vapor formulations in the dispensing interface 24 through thermal conduction. Alternatively, heat from the heating element 25 may be conducted to the one or more pre-vapor formulations by a heated conductive element or the heating element 25 may transfer heat to the incoming ambient air that is drawn through the e-vaping device 60 during vaping. The heated ambient air may heat the pre-vapor formulation by convection.

[0071] The dispensing interface 24 is configured to draw a pre-vapor formulation from the reservoir 23. The pre-vapor formulation drawn from the reservoir 23 into the dispensing interface 24 may be vaporized from the dispensing interface 24 based on heat generated by the heating element 25. During vaping, pre-vapor formulation may be transferred from the reservoir 23 and/or storage medium in the proximity of the heating element 25 through capillary action of the dispensing interface 24.

[0072] Still referring to FIG. 1A and FIG. 1B, in some example embodiments, the cartridge 70 includes a connector element 91. Connector element 91 may include one or more of a cathode connector element and an anode connector element. In the example embodiment illustrated in FIG. 1B, for example, electrical lead 26-1 is coupled to the connector element 91. As further shown in FIG. 1B, the connector element 91 is configured to couple with a power supply 12 included in the power supply section 72. If and/or when interfaces 74, 84 are coupled together, the connector element 91 and power supply 12 may be coupled together. Coupling connector element 91 and power supply 12 together may electrically couple lead 26-1 and power supply 12 together.

[0073] In some example embodiments, one or more of the interfaces 74, 84 include one or more of a cathode connector element and an anode connector element. In the example embodiment illustrated in FIG. 1B, for example, electrical lead 26-2 is coupled to the interface 74. As further shown in FIG. 1B, the power supply section 72 includes a lead 92 that couples the control circuitry 11 to the interface 84. If and/or when interfaces 74, 84 are coupled together, the coupled interfaces 74, 84 may electrically couple leads 26-2 and 92 together.

[0074] If and/or when interfaces 74, 84 are coupled together, one or more electrical circuits through the cartridge 70 and power supply section 72 may be established. The established electrical circuits may include at least the heating element 25, the control circuitry 11, and the power supply 12. The electrical circuit may include leads 26-1 and 26-2, lead 92, and interfaces 74, 84.

[0075] The connector element 91 may include an insulating material 91b and a conductive material 91a. The conductive material 91a may electrically couple lead 26-1 to power supply 12, and the insulating material 91b may insulate the conductive material 91a from the interface 74, such that a probability of an electrical short between the lead 26-1 and the interface 74 is reduced and/or prevented. For example, if and/or when the connector element 91 includes a cylindrical cross-section orthogonal to a longitudinal axis of the e-vaping device 60, the insulating material 91b

included in connector element 91 may be in an outer annular portion of the connector element 91 and the conductive material 91a may be in an inner cylindrical portion of the connector element 91, such that the insulating material 91b surrounds the conductive material 91a and reduces and/or prevents a probability of an electrical connection between the conductive material 91a and the interface 74.

[0076] Still referring to FIG. 1A and FIG. 1B, the power supply section 72 includes a sensor 13 responsive to air drawn into the power supply section 72 through an air inlet port 44a adjacent to a free end or tip end of the e-vaping device 60, a power supply 12, and control circuitry 11. In some example embodiments, including the example embodiment illustrated in FIG. 1B, the sensor 13 may be coupled to control circuitry 11 through lead 98. In some example embodiments, the sensor 13 may be coupled to control circuitry 11 through the electrode element 96 and the power supply 12. The power supply 12 may include a rechargeable battery. The sensor 13 may be one or more of a pressure sensor, a microelectromechanical system (MEMS) sensor, etc.

[0077] In some example embodiments, the power supply 12 includes a battery arranged in the e-vaping device 60 such that the anode is downstream of the cathode. A connector element 91 contacts the downstream end of the battery. The heating element 25 is coupled to the power supply 12 by at least the two spaced apart electrical leads 26-1 to 26-2.

[0078] The power supply 12 may be a Lithium-ion battery or one of its variants, for example a Lithium-ion polymer battery. Alternatively, the power supply 12 may be a nickel-metal hydride battery, a nickel cadmium battery, a lithium-manganese battery, a lithium-cobalt battery or a fuel cell. The e-vaping device 60 may be usable by an adult vaper until the energy in the power supply 12 is depleted or in the case of lithium polymer battery, a minimum voltage cut-off level is achieved. Further, the power supply 12 may be rechargeable and may include circuitry configured to allow the battery to be chargeable by an external charging device. To recharge the e-vaping device 60, a Universal Serial Bus (USB) charger or other suitable charger assembly may be used.

[0079] Still referring to FIG. 1A and FIG. 1B, upon completing the connection between the cartridge 70 and the power supply section 72, the power supply 12 may be electrically connected with the heating element 25 of the cartridge 70 upon actuation of the sensor 13. Air is drawn primarily into the cartridge 70 through one or more air inlet ports 44. The one or more air inlet ports 44 may be located along the outer housing 16 or at one or more of the interfaces 74, 84.

[0080] The sensor 13 may be configured to sense an air pressure drop and initiate application of voltage from the power supply 12 to the heating element 25. As shown in the example embodiment illustrated in FIG. 1B, some example embodiments of the power supply section 72 include a heater activation light 48 configured to glow when the heating element 25 is activated. The heater activation light 48 may include a light emitting diode (LED). Moreover, the heater activation light 48 may be arranged to be visible to an adult vaper during vaping. In addition, the heater activation light 48 may be utilized for e-vaping system diagnostics or to indicate that recharging is in progress. The heater activation light 48 may also be configured such that the adult vaper may activate and/or deactivate the heater activation

light 48 for privacy. As shown in FIG. 1A and FIG. 1B, the heater activation light 48 may be located on the tip end of the e-vaping device 60. In some example embodiments, the heater activation light 48 may be located on a side portion of the outer housing 17.

[0081] In addition, the at least one air inlet port 44a may be located adjacent to the sensor 13, such that the sensor 13 may sense air flow indicative of vapor being drawn through the outlet end of the e-vaping device 60. The sensor 13 may activate the power supply 12 and the heater activation light 48 to indicate that the heating element 25 is activated.

[0082] In some example embodiments, the control circuitry 11 may control the supply of electrical power to the heating element 25 responsive to the sensor 13. In some example embodiments, the control circuitry 11 is configured to adjustably control the electrical power supplied to the heating element 25. Adjustably controlling the supply of electrical power may include controlling the supply of electrical power such that supplied electrical power has a determined set of characteristics, where the determined set of characteristics may be adjusted. To adjustably control the supply of electrical power, the control circuitry 11 may control the supply of electrical power such that electrical power having one or more characteristics determined by the control circuitry 11 is supplied to the heating element 25. Such one or more selected characteristics may include one or more of voltage and current of the electrical power. Such one or more selected characteristics may include a magnitude of the electrical power. It will be understood that adjustably controlling the supply of electrical power may include determining a set of characteristics of electrical power and controlling the supply of electrical power such that electrical power supplied to the heating element 25 has the determined set of characteristics.

[0083] In some example embodiments, the control circuitry 11 may include a maximum, time-period limiter. In some example embodiments, the control circuitry 11 may include a manually operable switch for an adult vaper to initiate a vaping. The time-period of the electric current supply to the heating element 25 may be given, or alternatively pre-set (e.g., prior to controlling the supply of electrical power to the heating element 25), depending on the amount of pre-vapor formulation desired to be vaporized. In some example embodiments, the control circuitry 11 may control the supply of electrical power to the heating element 25 as long as the sensor 13 detects a pressure drop.

[0084] To control the supply of electrical power to heating element 25, the control circuitry 11 may execute one or more instances of computer-executable program code. The control circuitry 11 may include a processor and a memory. The memory may be a computer-readable storage medium storing computer-executable code.

[0085] The control circuitry 11 may include processing circuitry including, but not limited to, a processor, Central Processing Unit (CPU), a controller, an arithmetic logic unit (ALU), a digital signal processor, a microcomputer, a field programmable gate array (FPGA), a System-on-Chip (SoC), a programmable logic unit, a microprocessor, or any other device capable of responding to and executing instructions in a defined manner. In some example embodiments, the control circuitry 11 may be at least one of an application-specific integrated circuit (ASIC) and an ASIC chip.

[0086] The control circuitry 11 may be configured as a special purpose machine by executing computer-readable

program code stored on a storage device. The program code may include program or computer-readable instructions, software elements, software modules, data files, data structures, and/or the like, capable of being implemented by one or more hardware devices, such as one or more of the control circuitry mentioned above. Examples of program code include both machine code produced by a compiler and higher level program code that is executed using an interpreter.

[0087] The control circuitry **11** may include one or more electronic storage devices. The one or more storage devices may be tangible or non-transitory computer-readable storage media, such as random access memory (RAM), read only memory (ROM), a permanent mass storage device (such as a disk drive), solid state (e.g., NAND flash) device, and/or any other like data storage mechanism capable of storing and recording data. The one or more storage devices may be configured to store computer programs, program code, instructions, or some combination thereof, for one or more operating systems and/or for implementing the example embodiments described herein. The computer programs, program code, instructions, or some combination thereof, may also be loaded from a separate computer readable storage medium into the one or more storage devices and/or one or more computer processing devices using a drive mechanism. Such separate computer readable storage medium may include a USB flash drive, a memory stick, a Blu-ray/DVD/CD-ROM drive, a memory card, and/or other like computer readable storage media. The computer programs, program code, instructions, or some combination thereof, may be loaded into the one or more storage devices and/or the one or more computer processing devices from a remote data storage device through a network interface, rather than through a local computer readable storage medium. Additionally, the computer programs, program code, instructions, or some combination thereof, may be loaded into the one or more storage devices and/or the one or more processors from a remote computing system that is configured to transfer and/or distribute the computer programs, program code, instructions, or some combination thereof, over a network. The remote computing system may transfer and/or distribute the computer programs, program code, instructions, or some combination thereof, through a wired interface, an air interface, and/or any other like medium.

[0088] The control circuitry **11** may be a special purpose machine configured to execute the computer-executable code to control the supply of electrical power to heating element **25**. In some example embodiments, an instance of computer-executable code, when executed by the control circuitry **11**, causes the control circuitry **11** to control the supply of electrical power to heating element **25** according to an activation sequence. Controlling the supply of electrical power to heating element **25** may be referred to herein interchangeably as activating the one or more heating element **25**.

[0089] Still referring to FIG. 1A and FIG. 1B, when activated, the heating element **25** may heat a portion of the dispensing interface **24** surrounded by the heating element **25** for less than about 10 seconds. Thus, the power cycle (or maximum vaping length) may range in period from about 2 seconds to about 10 seconds (e.g., about 3 seconds to about 9 seconds, about 4 seconds to about 8 seconds or about 5 seconds to about 7 seconds).

[0090] In some example embodiments, the heating element **25** is electrically coupled to the control circuitry **11**. The control circuitry **11** may adjustably control the supply of electrical power to the heating element **25** to control an amount of heat generated by the heating element **25**.

[0091] In some example embodiments, the dispensing interface **24** includes an absorbent material, the absorbent material being arranged in fluidic communication with the heating element **25**. The absorbent material may include a wick having an elongated form and arranged in fluidic communication with the reservoir **23**. The wick may include a wicking material. The wicking material may be a fibrous wicking material. The wicking material may extend into reservoir **23**.

[0092] The pre-vapor formulation may include nicotine or may exclude nicotine. The pre-vapor formulation may include one or more tobacco flavors. The pre-vapor formulation may include one or more flavors that are separate from one or more tobacco flavors.

[0093] In some example embodiments, a pre-vapor formulation that includes nicotine may also include one or more acids. The one or more acids may be one or more of pyruvic acid, formic acid, oxalic acid, glycolic acid, acetic acid, isovaleric acid, valeric acid, propionic acid, octanoic acid, lactic acid, levulinic acid, sorbic acid, malic acid, tartaric acid, succinic acid, citric acid, benzoic acid, oleic acid, aconitic acid, butyric acid, cinnamic acid, decanoic acid, 3,7-dimethyl-6-octenoic acid, 1-glutamic acid, heptanoic acid, hexanoic acid, 3-hexenoic acid, trans-2-hexenoic acid, isobutyric acid, lauric acid, 2-methylbutyric acid, 2-methylvaleric acid, myristic acid, nonanoic acid, palmitic acid, 4-penenoic acid, phenylacetic acid, 3-phenylpropionic acid, hydrochloric acid, phosphoric acid, sulfuric acid and combinations thereof.

[0094] The storage medium of one or more reservoirs **23** may be a fibrous material including at least one of cotton, polyethylene, polyester, rayon and combinations thereof. The fibers may have a diameter ranging in size from about 6 microns to about 15 microns (e.g., about 8 microns to about 12 microns or about 9 microns to about 11 microns). The storage medium may be a sintered, porous or foamed material. Also, the fibers may be sized to be irrespirable and may have a cross-section that has a Y-shape, cross shape, clover shape or any other suitable shape. In some example embodiments, one or more reservoirs **23** may include a filled tank lacking any storage medium and containing only pre-vapor formulation.

[0095] Still referring to FIG. 1A and FIG. 1B, the reservoir **23** may be sized and configured to hold enough pre-vapor formulation such that the e-vaping device **60** may be configured for vaping for at least about 200 seconds. The e-vaping device **60** may be configured to allow each vaping to last a maximum of about 5 seconds.

[0096] The dispensing interface **24** may include a wicking material that includes filaments (or threads) having a capacity to draw one or more pre-vapor formulations. For example, a dispensing interface **24** may be a bundle of glass (or ceramic) filaments, a bundle including a group of windings of glass filaments, etc., all of which arrangements may be capable of drawing pre-vapor formulation through capillary action by interstitial spacings between the filaments. The filaments may be generally aligned in a direction perpendicular (transverse) or substantially perpendicular to the longitudinal direction of the e-vaping device **60**. In some

example embodiments, the dispensing interface **24** may include one to eight filament strands, each strand comprising a plurality of glass filaments twisted together. The end portions of the dispensing interface **24** may be flexible and foldable into the confines of one or more reservoirs **23**. The filaments may have a cross-section that is generally cross-shaped, clover-shaped, Y-shaped, or in any other suitable shape.

[0097] The dispensing interface **24** may include any suitable material or combination of materials, also referred to herein as wicking materials. Examples of suitable materials may be, but not limited to, glass, ceramic- or graphite-based materials. The dispensing interface **24** may have any suitable capillary drawing action to accommodate pre-vapor formulations having different physical properties such as density, viscosity, surface tension and vapor pressure.

[0098] In some example embodiments, the heating element **25** may include a wire coil that at least partially surrounds the dispensing interface **24**. The wire coil may be referred to as a heating coil wire. The heating coil wire may be a metal wire and/or the heating coil wire may extend fully or partially along the length of the dispensing interface **24**. The heating coil wire may further extend fully or partially around the circumference of the dispensing interface **24**. In some example embodiments, the heating coil wire may or may not be in contact with the outer surface of the dispensing interface **24**.

[0099] The heating element **25** may be formed of any suitable electrically resistive materials. Examples of suitable electrically resistive materials may include, but not limited to, titanium, zirconium, tantalum and metals from the platinum group. Examples of suitable metal alloys include, but not limited to, stainless steel, nickel, cobalt, chromium, aluminum-titanium-zirconium, hafnium, niobium, molybdenum, tantalum, tungsten, tin, gallium, manganese and iron-containing alloys, and super-alloys based on nickel, iron, cobalt, stainless steel. For example, the heating element **25** may be formed of nickel aluminate, a material with a layer of alumina on the surface, iron aluminate and other composite materials, the electrically resistive material may optionally be embedded in, encapsulated or coated with an insulating material or vice-versa, depending on the kinetics of energy transfer and the external physicochemical properties required. The heating element **25** may include at least one material selected from the group consisting of stainless steel, copper, copper alloys, nickel-chromium alloys, super alloys and combinations thereof. In some example embodiments, the heating element **25** may be formed of nickel-chromium alloys or iron-chromium alloys. In some example embodiments, the heating element **25** may be a ceramic heater having an electrically resistive layer on an outside surface thereof.

[0100] The dispensing interface **24** may extend transversely across the central channel **30** between opposing portions of the reservoir **23**. In some example embodiments, the dispensing interface **24** may extend parallel or substantially parallel to a longitudinal axis of the central channel **30**. In some example embodiments, including the example embodiment illustrated in FIG. 1B, the dispensing interface **24** may extend orthogonally or substantially orthogonally to the longitudinal axis of the central channel **30**.

[0101] In some example embodiments, the heating element **25** is a porous material that incorporates a resistance

heater formed of a material having a relatively high electrical resistance capable of generating heat relatively quickly.

[0102] In some example embodiments, the cartridge **70** may be replaceable. In other words, once the pre-vapor formulation of the cartridge **70** is depleted, only the cartridge **70** need be replaced. In some example embodiments, the entire e-vaping device **60** may be disposed once the reservoir **23** is depleted.

[0103] In some example embodiments, the e-vaping device **60** may be about 80 mm to about 110 mm long and about 7 mm to about 8 mm in diameter. For example, the e-vaping device **60** may be about 84 mm long and may have a diameter of about 7.8 mm.

[0104] FIG. 2A is a cross-sectional view of a vapor generator **22** that includes a superabsorbent polymer **202** within a tube **32**, according to some example embodiments. FIG. 2B is a cross-sectional view of a vapor generator **22** that includes a superabsorbent polymer **202** within at least a portion of the outer housing **16**, according to some example embodiments. FIG. 2C is a cross-sectional view of a vapor generator **22** that includes a superabsorbent layer **210** on an inner surface of at least a portion of the outer housing **16**, according to some example embodiments. FIG. 2D is a cross-sectional view of a vapor generator **22** that includes a superabsorbent layer **210** on an outer surface of at least a portion of tube **32**, according to some example embodiments. FIG. 2E is a cross-sectional view of a vapor generator **22** that includes a superabsorbent compartment **230**, according to some example embodiments. FIG. 2F is a cross-sectional view of a vapor generator **22** that includes a superabsorbent polymer **202** within a storage medium in the reservoir **23**, according to some example embodiments. FIG. 2G is a cross-sectional view of a vapor generator **22** that includes a reservoir **23** holding a formulation mixture **250** of pre-vapor formulation **252** and superabsorbent polymer **202**, according to some example embodiments. The vapor generator **22** shown in FIG. 1B may be any of the vapor generators **22** shown in FIGS. 2A-G.

[0105] Referring to FIG. 2A and FIG. 2B, in some example embodiments, a superabsorbent polymer **202** may be included in an interior of one or more of the surfaces defining the reservoir **23**. As shown in FIG. 2A, for example, a superabsorbent polymer **202** may be included in the tube **32**. As further shown in FIG. 2B, a superabsorbent polymer **202** may be included in a portion of the outer housing **16** defining at least one boundary of the reservoir **23** interior.

[0106] In some example embodiments, the superabsorbent polymer **202** may be in the form of a powder, as dispersed particles or fibers, as part of a coating layer, as part of a deposited layer, some combination thereof, or the like.

[0107] A superabsorbent polymer **202** included in a surface may be included within an interior of the material comprising the surface. For example, the superabsorbent polymer **202** may be injected, impregnated, etc. within the material comprising the tube **32** in FIG. 2A. In another example, the superabsorbent polymer **202** may be injected, impregnated, etc. within the material comprising the outer housing **16**, as shown in FIG. 2B.

[0108] In some example embodiments, the superabsorbent polymer **202** may be combined with one or more materials to collectively comprise the material of a surface. For example, the superabsorbent polymer **202** may be a fibrous material, and the tube **32** may be comprised of one or more fibrous materials. The fibrous superabsorbent polymer **202**

may be interwoven with the one or more fibrous materials to form the tube **32**. Such a tube **32** may include interwoven fibrous superabsorbent polymer **202** that is exposed to the interior of reservoir **23**.

[0109] Referring to FIG. 2C and FIG. 2D, in some example embodiments, a superabsorbent polymer **202** may be included in a superabsorbent layer **210**. The superabsorbent layer **210** may be on one or more of the surfaces defining an interior of the reservoir **23**.

[0110] As shown in FIGS. 2C-D, for example, the superabsorbent polymer **202** may be included in one or more superabsorbent layers **210**. As shown in FIGS. 2C-2D, the superabsorbent layers **210** each include one or more superabsorbent polymers **202** and one or more binder materials **204**. A superabsorbent layer **210** may be a mixture of superabsorbent polymer **202** and binder material **204**. The one or more binder materials **204** may include one or more inert or substantially inert materials, relative to at least one of the superabsorbent polymer **202**, a pre-vapor formulation that may be held in the reservoir **23**, and one or more elements comprising the vapor generator **22**. In some example embodiments, the binder material **204** includes a polyurethane binder material.

[0111] In some example embodiments, the mixture is a substantially uniform, homogenous mixture. In some example embodiments, a superabsorbent layer **210** includes a superabsorbent polymer **202** included in a superabsorbent structure, such that the mixture of a binder material **204** and one or more superabsorbent structures is heterogeneous. For example, a superabsorbent layer **210** may include one or more superabsorbent polymer **202** structures. Such structures may include spherical or substantially spherical capsules. The binder material **204** may bind the superabsorbent

[0113] In some example embodiments, the superabsorbent polymer **202** may be configured to absorb an amount of free water from a given mass of pre-vapor formulation. The amount of free water absorbed from the given mass of pre-vapor formulation may be equal to or greater than 10% of the given mass. For example, if and/or when the reservoir **23** holds 400 mg of pre-vapor formulation, the superabsorbent polymer **202** included in the reservoir **23** may be configured to absorb 40 mg of free water (also referred to herein as “moisture”) from the 400 mg of pre-vapor formulation.

[0114] In some example embodiments a structure of one or more superabsorbent polymers **202** may expand in volume by a proportion of the initial volume of the superabsorbent structure. For example, if and/or when the superabsorbent polymer **202** is included in a spherical capsule structure, the spherical capsule structure may expand to approximately 125% of the initial capsule volume upon the superabsorbent polymer **202** absorbing up to a maximum amount of free water.

[0115] Referring now to Table 1 below, the amount of water that may be absorbed by the superabsorbent layer **210** may be based on the dimensions of the reservoir **23** in which the layer **210** is included, the dimensions of the superabsorbent layer **210**, and the proportion of the superabsorbent layer **210** that is occupied by the superabsorbent polymer **202**. In Table 1, the superabsorbent layer **210** is on an inner surface **16a** of the outer housing **16** defining the outer boundary of a cylindrical reservoir **23**, and the superabsorbent polymer **202** occupies approximately 40% of the volume of the superabsorbent layer **210**.

TABLE 1

Reservoir inner diameter (cm)	Reservoir Volume (cm ³)	Reservoir Inner Surface Area (cm ²)	Layer Thickness (microns)	Layer Volume (cm ³)	Super-absorbent Volume (cm ³)	Water absorbed (mg)
0.50000	1.17810	4.71239	10.00000	0.00471	0.00188	1.88496
0.50000	1.17810	4.71239	50.00000	0.02356	0.00942	9.42478
0.50000	1.17810	4.71239	100.00000	0.04712	0.01885	18.84956
0.50000	1.17810	4.71239	200.00000	0.09425	0.03770	37.69911
0.80000	3.01593	7.53982	10.00000	0.00754	0.00302	3.01593
0.80000	3.01593	7.53982	50.00000	0.03770	0.01508	15.07964
0.80000	3.01593	7.53982	100.00000	0.07540	0.03016	30.15929
0.80000	3.01593	7.53982	200.00000	0.15080	0.06032	60.31858
1.00000	4.71239	9.42478	10.00000	0.00942	0.00377	3.76991
1.00000	4.71239	9.42478	50.00000	0.04712	0.01885	18.84956
1.00000	4.71239	9.42478	100.00000	0.09425	0.03770	37.69911
1.00000	4.71239	9.42478	200.00000	0.18850	0.07540	75.39822

polymer **202** capsules to the surface **16a**, **32a** on which the superabsorbent layer **210** is located, respectively.

[0112] As shown in FIG. 2C, the superabsorbent layer **210** may be on an inner surface **16a** of the outer housing **16**. As shown in FIG. 2D, the superabsorbent layer **210** may be on an outer surface **32a** of the tube **32**. Each of the superabsorbent layers **210** shown in FIGS. 2C-D is on one or more surfaces that are exposed to the reservoir **23** interior. The superabsorbent layers **210** are thus each in fluid communication with an interior of a reservoir **23** that is at least partially defined by the surface on which the superabsorbent layers **210** are located, respectively.

[0116] Referring to FIG. 2E, in some example embodiments, reservoir **23** may include a formulation compartment **220** and a superabsorbent compartment **230**. The formulation compartment **220** may be configured to hold the pre-vapor formulation therein. The formulation compartment **220** may include a storage medium that may hold pre-vapor formulation. The superabsorbent compartment **230** may include superabsorbent structures **234** and a partition **232** that defines at least one boundary of the superabsorbent compartment **230**.

[0117] The superabsorbent structures **234** included in the superabsorbent compartment **230** may include one or more

superabsorbent polymers **202**. The superabsorbent structures **234** may include one or more types of structures **234**. The one or more types of structures **234** may include one or more capsule structures. The one or more structures may be spherical or substantially spherical.

[0118] In some example embodiments, the partition **232** may be selectively permeable. The selectively-permeable partition **232** may enable pre-vapor formulation circulation between the interiors of the formulation compartment **220** and the superabsorbent compartment **230**, respectively. The selectively permeable partition **232** may restrict the superabsorbent structures **234** to the interior of the superabsorbent compartment **230**, thereby reducing and/or preventing the superabsorbent polymer **202** from crossing the partition **232** into the formulation compartment **220**.

[0119] Based on enabling pre-vapor formulation circulation into the superabsorbent compartment **230** while restricting the superabsorbent polymer **202** to the superabsorbent compartment **230** interior, the partition **232** may enable the pre-vapor formulation to circulate in and out of the superabsorbent compartment **230** so that the superabsorbent structures **234** held in the superabsorbent compartment **230** may remove free water from the circulating pre-vapor formulation.

[0120] In the example embodiments illustrated in FIG. 2E, the superabsorbent compartment **230** is located at an end of the reservoir **23** such that the partition **232**, outer housing **16**, tube **32**, and gasket **14** collectively define the superabsorbent compartment **230** interior. In some example embodiments, the superabsorbent compartment **230** may be located at one or more additional locations in the reservoir **23**. For example, the superabsorbent compartment **230** may be located at a midpoint in the reservoir **23**, such that separate formulation compartments are located on opposite ends of the superabsorbent compartment **230**.

[0121] In some example embodiments, the partition **232** may be a containment structure that at least partially encloses the superabsorbent polymer **202** to form a superabsorbent compartment **230** in the reservoir **23**. For example, the partition **232** may be a bag structure that encloses one or more superabsorbent structures **234** and is located within the reservoir **23**. Such a compartment **230** formed by a bag structure enclosing one or more superabsorbent structures **234** may be enclosed within the formulation compartment **220**.

[0122] The partition **232** may include one or more instances of mesh material. For example, the partition **232** may include a material that has one or more holes. The one or more holes may be sufficiently small to restrict superabsorbent structures **234** from passing through the partition **232** and sufficiently large to direct pre-vapor formulation through the partition **232** to circulate between the compartments **220**, **230**. As a result, the partition **232** may reduce and/or prevent superabsorbent structures **234** from approaching within a certain margin proximity to the vaporizer assembly **28**. Reducing and/or preventing superabsorbent structures **234** from approaching within a certain margin proximity to the vaporizer assembly **28** may reduce and/or prevent blockage of a fluid pathway between the reservoir **23** and the dispensing interface **24**. Reducing and/or preventing such blockage may thereby reduce and/or prevent superabsorbent structures **234** within the reservoir **23** from at least partially restricting pre-vapor formulation from being drawn into the dispensing interface **24**.

[0123] As shown in FIG. 2E, the vaporizer assembly **28** may be configured to draw pre-vapor formulation from the formulation compartment **220**. As shown, the dispensing interface **24** of the vaporizer assembly **28** is coupled to the formulation compartment **220**. The vaporizer assembly **28** may draw pre-vapor formulation that has circulated in and out of the superabsorbent compartment **230** at least once and therefore may be at least partially relieved of free water.

[0124] The superabsorbent structures **234** may form an absorbent bed within the superabsorbent compartment **230** interior. The superabsorbent structures **234** may be packed into the compartment interior according to a particular packing density. For example, the superabsorbent structures **234** in the compartment may form a superabsorbent bed having a packing density of approximately 0.7.

[0125] Referring now to Table 2 below, the amount of water that may be absorbed by the superabsorbent polymer **202** in a superabsorbent compartment **230** may be based on the packing density of the superabsorbent structures **234** in the superabsorbent compartment **230**, the density of the superabsorbent structures **234**, and the dimensions of the reservoir **23** in which the superabsorbent compartment **230** is included. In Table 2, the packing density of the superabsorbent structures **234** is approximately 0.7, the density of the superabsorbent structures **234** is approximately 1 g/cm³, the reservoir **23** is a cylindrical reservoir that is approximately 1 cm in length. The one or more superabsorbent polymers **202** in the superabsorbent structures **234** may be configured to absorb an amount of water that is equal to the mass of the superabsorbent polymer **202**. In some example embodiments, the one or more superabsorbent polymers **202** in the superabsorbent structures **234** may be configured to absorb an amount of water that is greater than the mass of the superabsorbent polymer **202**.

TABLE 2

Superabsorbent Compartment Length (cm)	Compartment Volume (cm ³)	Superabsorbent Structure Volume (cm ³)	Water Absorbed (mg)
0.25000	0.09817	0.06872	68.72234
0.50000	0.19635	0.13744	137.44468
0.75000	0.29452	0.20617	206.16702
1.00000	0.39270	0.27489	274.88936
0.25000	0.25133	0.17593	175.92919
0.50000	0.50265	0.35186	351.85838
0.75000	0.75398	0.52779	527.78757
1.00000	1.00531	0.70372	703.71675
0.25000	0.39270	0.27489	274.88936
0.50000	0.78540	0.54978	549.77871
0.75000	1.17810	0.82467	824.66807
1.00000	1.57080	1.09956	1099.55743

[0126] Referring to FIG. 2F, in some example embodiments the superabsorbent polymer **202** may be included within a storage medium **240** of the reservoir **23**. In some example embodiments, the one or more superabsorbent polymers **202** may be included within an interior of the storage medium **240**. For example, the superabsorbent polymer **202** may be injected, impregnated, etc. within the storage medium **240**. In some example embodiments, the superabsorbent polymer **202** may be a fibrous material that is interwoven with one or more materials comprising the storage medium **240**. In some example embodiments, the

superabsorbent polymer **202** is included in a layer (“coating”) on one or more outer surfaces of the storage medium material.

[0127] Referring to FIG. 2G, in some example embodiments, the superabsorbent polymer **202** is included in a formulation mixture **250** that further includes the pre-vapor formulation **252**. The formulation mixture **250** may be a liquid mixture. The formulation mixture **250** may be a uniform or substantially uniform mixture of the pre-vapor formulation **252** and the superabsorbent polymer **202**. The formulation mixture **250** may be a suspension of the superabsorbent polymer **202** in the pre-vapor formulation **252**. The formulation mixture **250** may be a suspension of the pre-vapor formulation **252** in the superabsorbent polymer **202**. The formulation mixture **250** may be added to the reservoir **23** subsequent to mixing the pre-vapor formulation **252** and the superabsorbent polymer **202** together to establish the formulation mixture **250**. In some example embodiments, the formulation mixture **250** may be a mixture of 40% superabsorbent polymer **202** and 60% pre-vapor formulation **252** by volume. In some example embodiments, the formulation mixture **250** may be a mixture of 40% superabsorbent polymer **202** and 60% pre-vapor formulation **252** by mass.

[0128] In the example embodiments illustrated in FIG. 2G, the pre-vapor formulation **252** in the formulation mixture **250** is represented by hollow circles and the superabsorbent polymer **202** in the formulation mixture **250** is represented by filled circles. It will be understood that the illustrated representations **250**, **252** are representative of a mixture of the pre-vapor formulation **252** and the superabsorbent polymer **202** in the formulation mixture **250** and are not intended to indicate that the pre-vapor formulation **252** and superabsorbent polymer **202** are each quantized particles, elements, etc. within a mixture **250**. For example, in some example embodiments, the pre-vapor formulation **252** and the superabsorbent polymer **202** may both be liquids, such that the formulation mixture **250** is a solution of at least pre-vapor formulation **252** and superabsorbent polymer **202**.

[0129] In some example embodiments, the formulation mixture **250** may include a pre-vapor formulation **252** and superabsorbent polymer **202** that are in different phases, respectively. For example, the formulation mixture **250** may include a dispersion, a suspension, some combination thereof, or the like. The formulation mixture **250** may include a dispersion, suspension, etc. of superabsorbent polymer **202** gel particles in a pre-vapor formulation. The formulation mixture **250** may include a dispersion, suspension, etc. of superabsorbent polymer **202** solid particles in a pre-vapor formulation, some combination thereof, or the like. In some example embodiments, the pre-vapor formulation **252** may be in a phase that is different from a liquid phase. For example, the pre-vapor formulation **252** may be a liquid, a gel material, a solid material, some combination thereof, or the like.

[0130] FIG. 3A, FIG. 3B, and FIG. 3C illustrate a method of at least partially forming a cartridge **70** that includes a superabsorbent polymer **202** on a surface that at least partially defines an interior of a reservoir **23** of the cartridge **70**, according to some example embodiments.

[0131] In some example embodiments, a superabsorbent polymer **202** may be included in a superabsorbent layer **210** that is applied to one or more surfaces. The one or more surfaces may at least partially define an interior of the

cartridge **70**, such that the one or more surfaces at least partially define a reservoir in the cartridge.

[0132] Referring back to FIGS. 1A-B, the interior of the cartridge **70** may be at least partially defined by the outer housing **16**. The outer housing **16**, as shown in FIG. 1A-B, may be a cylinder that defines a cylindrical interior of the cartridge **70**. At least a portion of the inner surface **16a** of the outer housing **16** may define an outer boundary of the reservoir **23**.

[0133] Referring back to FIGS. 3A-C, the housing **16** may include a material **302** that is manipulated **306** from an initial shape into a shape that at least partially defines an interior of the cartridge **70**. In the example embodiments shown in FIGS. 3A-C, the outer housing **16** material **302** may initially be a flat surface having a planar initial shape. The material **302** may be manipulated **306** from a planar shape into a circular cylindrical final shape to form the outer housing **16**. It will be understood that the initial shape and final shape may be different from the shapes illustrated in FIGS. 3A-C. For example, the initial shape of the outer housing material **302** may be a curved surface and the final shape may be a triangular cylinder shape.

[0134] In the example embodiments shown in FIGS. 3A-C, a superabsorbent layer **210** may be formed on the outer housing material **302** during a process of manipulating the outer housing material **302** into a final shape to form the outer housing **16**. As shown, the superabsorbent layer **210** may be applied to a surface **16a** of the outer housing material **302** prior to manipulating the outer housing material **302** into a final shape to form the outer housing **16**.

[0135] As shown in FIG. 3A, the outer housing material **302** may be in an initial shape that is planar or substantially planar. The outer housing material **302** may have an exposed surface **16a**.

[0136] As shown in FIG. 3B, a superabsorbent layer **210** may be applied to the exposed surface **16a** so that the superabsorbent layer **210** is on the exposed surface **16a** of the outer housing material **302**. The superabsorbent layer **210** may be applied to the exposed surface **16a** by one or more applicators **304**.

[0137] In some example embodiments, the one or more applicators **304** may include a sprayer device that sprays the material comprising the superabsorbent layer **210** onto the exposed surface **16a** to form the superabsorbent layer **210** on the exposed surface **16a**.

[0138] In some example embodiments, the superabsorbent layer **210** may be applied to the surface **16a** by one or more various applicators **304**. In some example embodiments, the superabsorbent layer **210** may be applied to the surface **16a** via lamination, deposition, coating via a non-physical applicator (e.g., a sprayer), coating via a physical applicator (e.g., a brush device), some combination thereof, or the like.

[0139] As shown in FIG. 3C, the outer housing material **302** may be manipulated **306** from the initial shape as shown in FIG. 3A-B to a final shape that at least partially defines an exterior of the cartridge **70** and an interior of the cartridge **70**. The outer housing material **302** in the final shape may be referred to as the outer housing **16**.

[0140] Such manipulation **306** may be implemented via one or more various systems. The outer housing material **302** may be manipulated **306** into the final shape based on manipulation of the outer housing material **302** by one or more manipulators. The manipulator may be one or more of a human operator, a machine, some combination thereof, etc.

The machine may be a fabrication machine. The machine may be a special purpose machine configured to implement the manipulating based on executing program code stored in a memory device.

[0141] In the example embodiments shown in FIG. 3C, the outer housing material 302 is manipulated 306 into a circular cylinder final shape to form an outer housing 16 defining an interior space 310 that may at least partially comprise an interior of the cartridge 70 shown in FIG. 1B.

[0142] As shown in FIG. 3C, the outer housing material 302 is manipulated 306 so that the surface 16a defines an outer boundary of the interior space 310. Because the superabsorbent layer 210 is on the surface 16a, the superabsorbent layer 210 is exposed to the interior space 310 based on the outer housing material 302 being manipulated into the final shape.

[0143] In some example embodiments, the superabsorbent layer 210 may be applied to the surface 16a subsequent to the outer housing material 302 being manipulated 306 into the final shape. For example, the superabsorbent layer 210 may be applied to the surface 16a, subsequent to the outer housing 16 being formed via the manipulation of the outer housing material 302 as shown in FIG. 3C.

[0144] In some example embodiments, some or all of the process illustrated in FIGS. 3A-C may be implemented in the presence of a “dry” environment. In some example embodiments, a “dry” environment is an environment in which the relative humidity is less than 30%. In some example embodiments, a “dry” environment is an environment in which the relative humidity is 0%.

[0145] FIG. 4 is a flowchart illustrating a method for configuring a cartridge to absorb free water from a pre-vapor formulation held in the reservoir, according to some example embodiments. The configuring may be implemented with regard to any of the example embodiments of cartridges included herein. In some example embodiments, one or more portions of the configuring are implemented by a configurator. The configurator may be one or more of a human operator, a machine, some combination thereof, etc. The machine may be a fabrication machine. The machine may be a special purpose machine configured to implement the configuring based on executing program code stored in a memory device. The configurator may include the manipulator described above with reference to FIGS. 3A-C.

[0146] Referring to FIG. 4, in some example embodiments, one or more of 402-414 may be implemented. In some example embodiments, a selected one of 402-414 is implemented and a remainder one or more of 402-414 is not implemented.

[0147] At 402, the configurator incorporates a superabsorbent polymer into an outer housing material comprising at least a portion of the outer housing. The superabsorbent polymer may be incorporated into the outer housing material through at least one of impregnation into the outer housing material, injection into the outer housing material, some combination thereof, or the like. The superabsorbent polymer may be incorporated into the outer housing material prior to the outer housing material being manipulated to form the outer housing.

[0148] At 404, the configurator incorporates a superabsorbent polymer into at least a portion of the inner tube material. The superabsorbent polymer may be incorporated into the outer housing material through at least one of impregnation into the inner tube material, injection into the

inner tube material, some combination thereof, or the like. The superabsorbent polymer may be incorporated into the inner tube material prior to the inner tube material being manipulated to form the inner tube.

[0149] At 406, the configurator incorporates a superabsorbent polymer into at least a portion of a storage medium material. The superabsorbent polymer may be incorporated into the storage medium material through at least one of impregnation into the storage medium material, injection into the storage medium material, some combination thereof, or the like. The superabsorbent polymer may be incorporated into the storage medium material prior to the storage medium material being manipulated to form the storage medium.

[0150] At 408, the configurator applies a superabsorbent layer to a surface of the inner tube material. The superabsorbent layer may include at least one superabsorbent polymer and at least one binder material. The superabsorbent polymer may be included in one or more superabsorbent structures. The superabsorbent layer may be applied to the surface by one or more applicators. The superabsorbent layer may be applied to a surface of the inner tube material prior or subsequent to the inner tube material being manipulated to form the inner tube.

[0151] At 410, the configurator applies a superabsorbent layer to a surface of the outer housing material. The superabsorbent layer may include at least one superabsorbent polymer and at least one binder material. The superabsorbent polymer may be included in one or more superabsorbent structures. The superabsorbent layer may be applied to the surface by one or more applicators. The superabsorbent layer may be applied to a surface of the outer housing material prior or subsequent to the outer housing material being manipulated to form the outer housing.

[0152] At 420, the configurator inserts the inner tube into an interior space defined by the outer housing to at least partially define the reservoir between an outer surface of the inner tube and an inner surface of the outer housing.

[0153] At 430, the configurator adds the storage medium into the interior of the reservoir.

[0154] At 412, the configurator forms a superabsorbent compartment in at least a portion of the reservoir interior. Forming a superabsorbent compartment in a portion of the reservoir interior may define a remainder of the reservoir interior as a formulation compartment of the reservoir. The superabsorbent compartment may include one or more superabsorbent structures and a selectively permeable partition that contains the superabsorbent structures within an interior of the superabsorbent compartment and enables pre-vapor formulation circulation through the partition between the interior of the superabsorbent compartment and one or more formulation compartments external to the superabsorbent compartment.

[0155] At 414, the configurator mixes at least one superabsorbent polymer with at least one pre-vapor formulation to form a formulation mixture. The at least one superabsorbent polymer may be included in one or more superabsorbent structures included in the formulation mixture. The formulation mixture may include a suspension, a homogenous mixture, a heterogeneous mixture, a solution, a colloid, some combination thereof, or the like.

[0156] At 440, the configurator adds at least the pre-vapor formulation to the reservoir. In some example embodiments, where 414 is implemented such that a formulation mixture

including the pre-vapor formulation and the superabsorbent polymer is formed, the configurator may add the formulation mixture to the reservoir at 440.

[0157] While a number of example embodiments have been disclosed herein, it should be understood that other variations may be possible. Such variations are not to be regarded as a departure from the spirit and scope of the present disclosure, and all such modifications as would be obvious to one skilled in the art are intended to be included within the scope of the following claims.

We claim:

1. A method of forming a cartridge, comprising:
manipulating a housing material into a cylindrical shape to form an outer housing, an interior surface of the outer housing defining an interior void;
applying a polymer to a surface of an inner tube, the polymer including a cross-linked polyacrylate copolymer;
inserting the inner tube into the interior void, the polymer being integral to or directly attached to the surface of the inner tube facing the interior void;
inserting a storage medium into the interior void between the inner tube and the outer housing, the storage medium being independent of the polymer; and
adding a pre-vapor formulation to the storage medium of the interior void, the polymer configured to absorb free water from the pre-vapor formulation and retain the free water, the polymer being substantially inert to the pre-vapor formulation.
2. The method of claim 1, wherein the applying the polymer includes incorporating the polymer into the inner tube.
3. The method of claim 1, wherein the manipulating the housing material includes incorporating the polymer into at least one surface of the housing material.

4. The method of claim 1, wherein the manipulating the housing material includes applying a layer of the polymer onto at least one surface of the housing material.

5. The method of claim 1, wherein the inserting the storage medium includes inserting a selectively permeable partition at a boundary of the polymer and the storage medium to form a storage medium compartment and an absorbent compartment.

6. A method of forming a cartridge, comprising:

manipulating a housing material into a cylindrical shape to form a housing, the cylindrical shape defining an interior void;

applying a polymer material to a surface of an inner tube, the polymer material including a cross-linked polyacrylate copolymer;

inserting the inner tube into the interior void, the polymer material being integral to or directly attached to the surface of the inner tube facing the interior void;

combining a storage medium with the polymer material to form an absorbent matrix;

inserting the absorbent matrix into the interior void; and
adding a pre-vapor formulation to the storage medium of the interior void, the polymer material configured to absorb free water from the pre-vapor formulation independent of the storage medium and retain the free water, the polymer material being substantially inert to the pre-vapor formulation.

7. The method of claim 6, wherein the applying the polymer material includes incorporating the polymer material into the inner tube.

8. The method of claim 6, wherein the manipulating the housing material includes incorporating the polymer material into at least one surface of the housing material.

9. The method of claim 6, wherein the manipulating the housing material includes applying a layer of the polymer material onto at least one surface of the housing material.

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